SpaceX Falcon 9

First Stage Landing Prediction

IBM Data Science Capstone Project

GitHub: [Your GitHub Repository Link]

Executive Summary

Predict if SpaceX Falcon 9 first stage will land successfully to determine launch costs and enable competitive bidding against SpaceX.

Key Findings

- Launch success rate has improved significantly over time
- KSC LC-39A has the highest success rate among launch sites
- Payload mass and orbit type correlate with landing outcomes
- Machine learning models achieved ~83%

Key Metrics

83%

Model Accuracy

~66%

Overall Success Rate

Introduction

SpaceX advertises Falcon 9 rocket launches at $62 million, while other providers cost upward of $165 million. The significant savings stem from SpaceX's ability to reuse the first stage.

Project Goal

If we can predict whether the first stage will land successfully, we can determine the cost of a launch. This information is valuable for alternate companies bidding against SpaceX for rocket launches.

Approach

1. Data Collection
SpaceX APIs & Web Scraping

2. Data Wrangling
Cleaning & Feature Engineering

3. Exploratory Analysis
SQL, Visualization, Folium Maps

4. Interactive Analytics
Plotly Dash Dashboard

5. Machine Learning
Classification Models

Data Collection & Wrangling Methodology

Data Collection

SpaceX REST API

- Collected launch records using SpaceX API
- Extracted booster versions, payloads, launch sites, orbits, and landing outcomes

Web Scraping

- Scraped Falcon 9 launch data from Wikipedia using BeautifulSoup
- Parsed HTML tables for historical records

Data Wrangling

Data Cleaning

- Handled missing values
- Filtered relevant columns
- Standardized data formats

Feature Engineering

- Created binary classification target: Class = 1 (successful landing), Class = 0 (failed landing)
- Identified bad outcomes: False Ocean, False RTLS, False ASDS, None ASDS, None None

Dataset: 90+ launch records from 2010-2020 | Success Rate: ~66%

EDA & Interactive Analytics Methodology

SQL Analysis

- Queried SQLite database
- Analyzed launch sites, orbits, payloads
- Aggregated success/failure rates
- Identified patterns in booster versions

Visualization (Matplotlib/Seaborn)

- Flight number vs. Launch site
- Payload vs. Launch site
- Success rate vs. Orbit type
- Yearly launch trend analysis

Folium Interactive Maps

- Mapped all launch sites geographically
- Visualized success/failure markers
- Calculated distances to proximities
- Analyzed geographical patterns

Plotly Dash Dashboard

- Interactive dropdown for launch sites
- Pie charts for success/failure counts
- Payload slider (0-10,000 kg)
- Scatter plots: payload vs. outcome

Predictive Analysis Methodology

Machine Learning Pipeline

Data Preparation
- Standardized features using StandardScaler
- Split data into training (80%) and testing (20%) sets

Models Evaluated
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree Classifier
- K-Nearest Neighbors (KNN)

Hyperparameter Tuning
- GridSearchCV for optimal parameters
- Cross-validation to prevent overfitting

Evaluation Metrics
✓ Accuracy

✓ Confusion Matrix

✓ Precision & Recall

✓ F1-Score

✓ Jaccard Score

EDA with Visualization - Key Results

Launch Success Trend

Success rate improved significantly from 2013 onwards, reaching 80%+ in recent years

Payload Mass Impact

Heavier payloads (>6,000 kg) show lower success rates. Lighter payloads have higher landing success

Orbit Type Patterns

LEO and ISS orbits show highest success rates. GTO and SSO orbits have more failures

Launch Site Performance

KSC LC-39A demonstrates the highest success rate among all launch sites

Key Insight: Booster version FT (Full Thrust) shows marked improvement in landing success

EDA with SQL - Key Findings

Launch Sites

- 4 unique launch sites: CCAFS SLC-40, VAFB SLC-4E, KSC LC-39A, CCAFS LC-40
- KSC LC-39A shows highest success percentage

Payload Analysis

- Total payload mass carried by NASA (CRS): 45,596 kg
- Average payload mass for F9 v1.1: ~2,928 kg

Landing Outcomes

- First successful ground pad landing: 2015-12-22
- Success outcomes outnumber failures in recent years
- ASDS (drone ship) landings show improving trend

Database Stats

90+
Total Records

10+
Orbit Types

8+
Booster Versions

Interactive Map Analysis - Folium

Geographical Insights

Launch Site Locations
- All 4 launch sites mapped with markers
- Coastal proximity for ocean recovery

Success/Failure Patterns
- Green markers: Successful landings
- Red markers: Failed landings
- Clustering of success at newer sites

Proximity Analysis
- Calculated distances to highways, cities
- Railway proximity for logistics
- Coastline access for drone ship landings

Launch Sites

1. CCAFS SLC-40
Cape Canaveral, FL
28.56° N, 80.58° W

2. KSC LC-39A
Kennedy Space Center, FL
28.61° N, 80.60° W

3. VAFB SLC-4E
Vandenberg AFB, CA
34.63° N, 120.61° W

4. CCAFS LC-40
Cape Canaveral, FL
28.56° N, 80.57° W

Plotly Dash Dashboard - Insights

Interactive Dashboard Features

Site-Specific Analysis

- Dropdown selection for individual launch sites or all sites
- Pie charts showing success vs. failure distribution
- KSC LC-39A showed highest success rate (76.9%)

Payload Impact

- Payload range slider: 0-10,000 kg
- Scatter plot: Payload mass vs. Landing outcome
- Color-coded by booster version (FT, v1.0, v1.1, B4, B5)
- Payload range 2,000-6,000 kg shows highest success

Key Findings

Best Launch Site:
KSC LC-39A

Optimal Payload:
2,000-6,000 kg

Best Booster:
FT & B5 versions

Success Rate:
~66% overall

Machine Learning Results

Model Performance Comparison

Model	Accuracy	Jaccard	F1-Score	Status
Decision Tree	83.3%	0.74	0.81	✓ Best
K-Nearest Neighbors	83.3%	0.74	0.80	Good
Support Vector Machine	83.3%	0.74	0.80	Good
Logistic Regression	83.3%	0.74	0.80	Good

Model Insights

- All models achieved 83.3% accuracy on test set
- Decision Tree selected as best model with highest F1-score (0.81)
- GridSearchCV optimized hyperparameters for each algorithm
- Confusion matrix shows strong true positive rate

Key Takeaways

Conclusion

1. Predictability Achieved

Machine learning models can predict Falcon 9 landing success with 83% accuracy

2. Critical Success Factors

Launch site (KSC LC-39A), payload range (2-6K kg), and booster version (FT/B5) significantly impact landing outcomes

3. Business Value

Companies can estimate launch costs and make informed competitive bids against SpaceX

Thank you for your attention!